NESDIS Commercial Data Program (CDP)

Industry Day

National Environmental Satellite, Data, and Information Service

April 9, 2026

NESDIS CDP Industry Day Agenda

1:00pm	Welcome and Introduction of NOAA Executive Leadership
1:05pm	Opening Remarks from Taylor Jordan, NOAA Assistant Secretary of Commerce for Environmental Observation and Prediction
1:15pm	Irene Parker, Acting NESDIS Assistant Administrator
1:25pm	Michael Conroy, NOAA Acquisition & Grants Office
1:30pm	Natalie Laudier, NESDIS Commercial Data Program Manager
1:55pm	Break
2:00pm	Question & Answer Session with CDP Panel

NOAA NESDIS Commercial Data Program Industry Day

Irene Parker, *NESDIS Assistant Administrator (acting)*

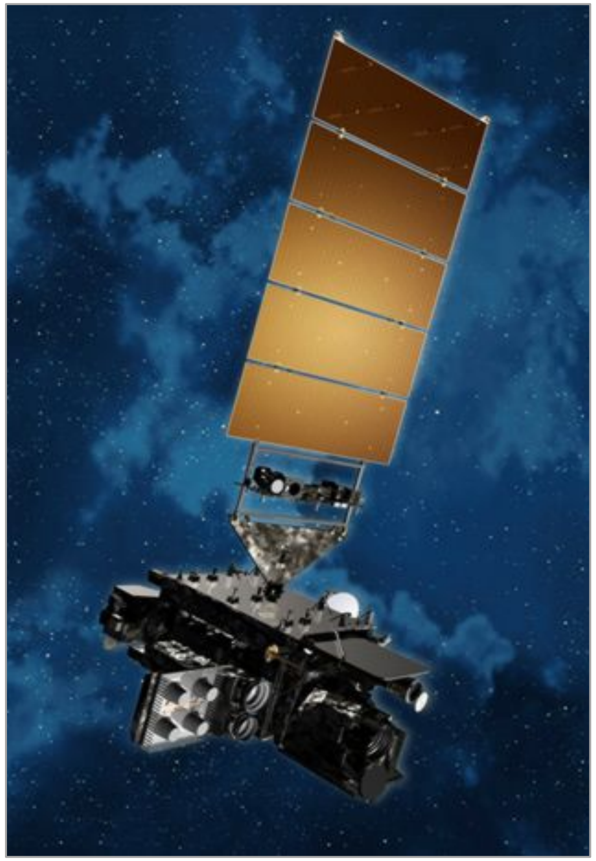
National Environmental Satellite, Data, and Information Service

April 9, 2026

NESDIS Mission

Essential to National Security, Safety, & Prosperity

- Operates the Nation's weather satellites, 24/7
- Acquires next-generation Earth observation satellites
- Provides data and imagery for predictive environmental and atmospheric modeling from our own satellites, our partners', and the commercial sector
- Maintains one of the most comprehensive archives of environmental data on Earth



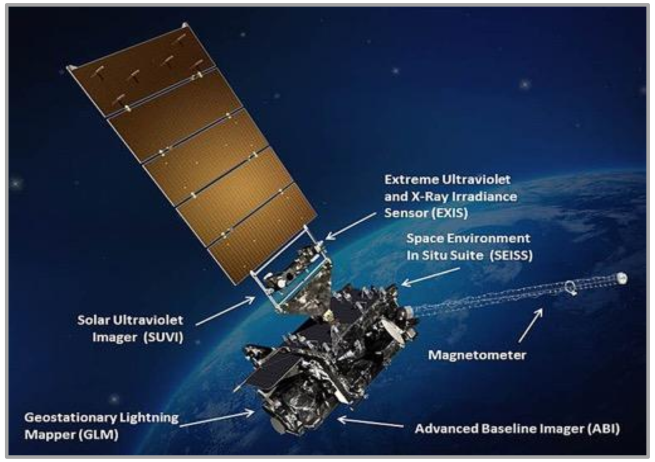
~90% of the data used in weather forecast models come from satellites

Extreme Weather Prediction Starts with NOAA Satellites

NOAA Observations: Buy, Partner, Build

Low Earth Satellites (LEO)

500 miles above Earth



Geostationary Satellites (GEO)

22,000 miles above Earth

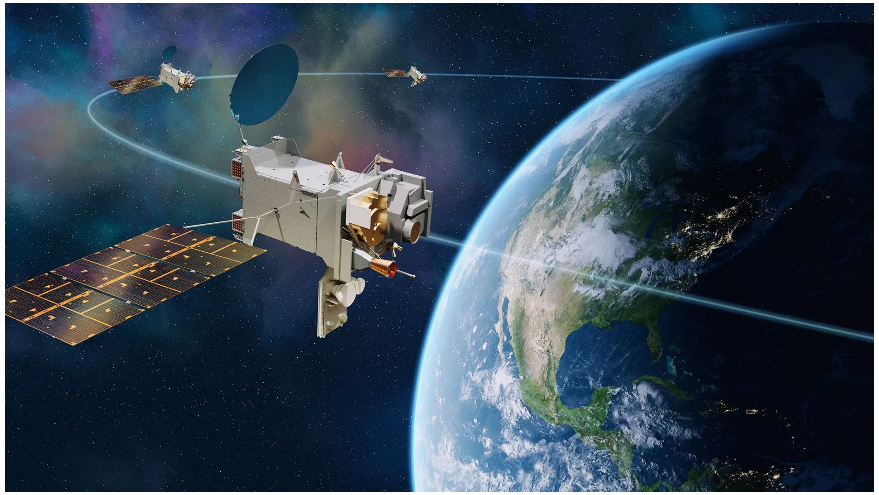
Space Weather Satellites

Lagrange 1: ~1 million miles from Earth and other observation points

NOAA relies on commercial weather data alongside government program and partner observations to ***improve coverage*** and ***achieve more accurate weather forecasts*** at the ***best value to the U.S. taxpayer.***

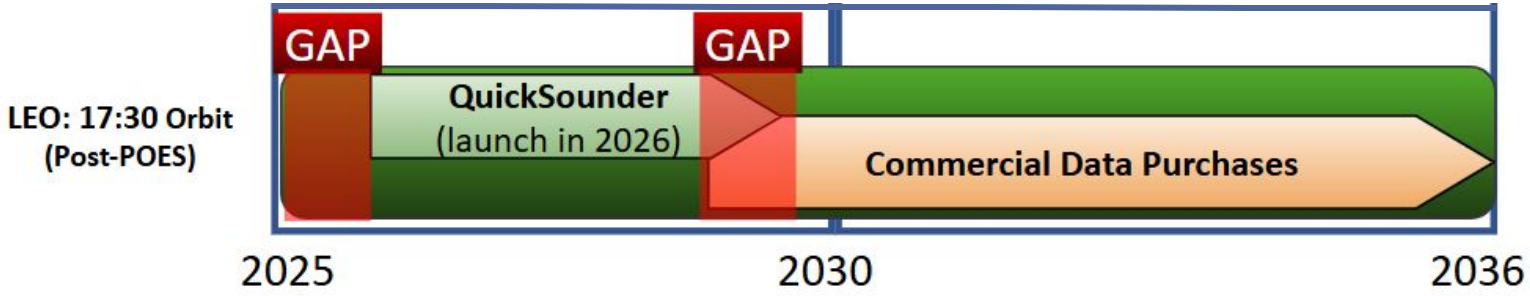
Modernizing NOAA Observing Systems

- Future NESDIS satellite programs will continue to tie to key weather observational requirements and advance the capabilities of NOAA satellites
  - GeoXO, NEON, and SW Next will fly advanced, next-gen instruments, building on the capabilities of the instruments currently flying on GOES-R, JPSS, and SWFO
  - Leverage commercial capabilities, and continue to foster interagency and international partnerships
- NESDIS will develop observations towards becoming a world leader in NWP computing
  - Advancing global and regional models, hurricane modeling, and severe weather prediction.


GeoXO


QuickSounder

Near-Term Priority: Prevent Gaps for Critical Continuity

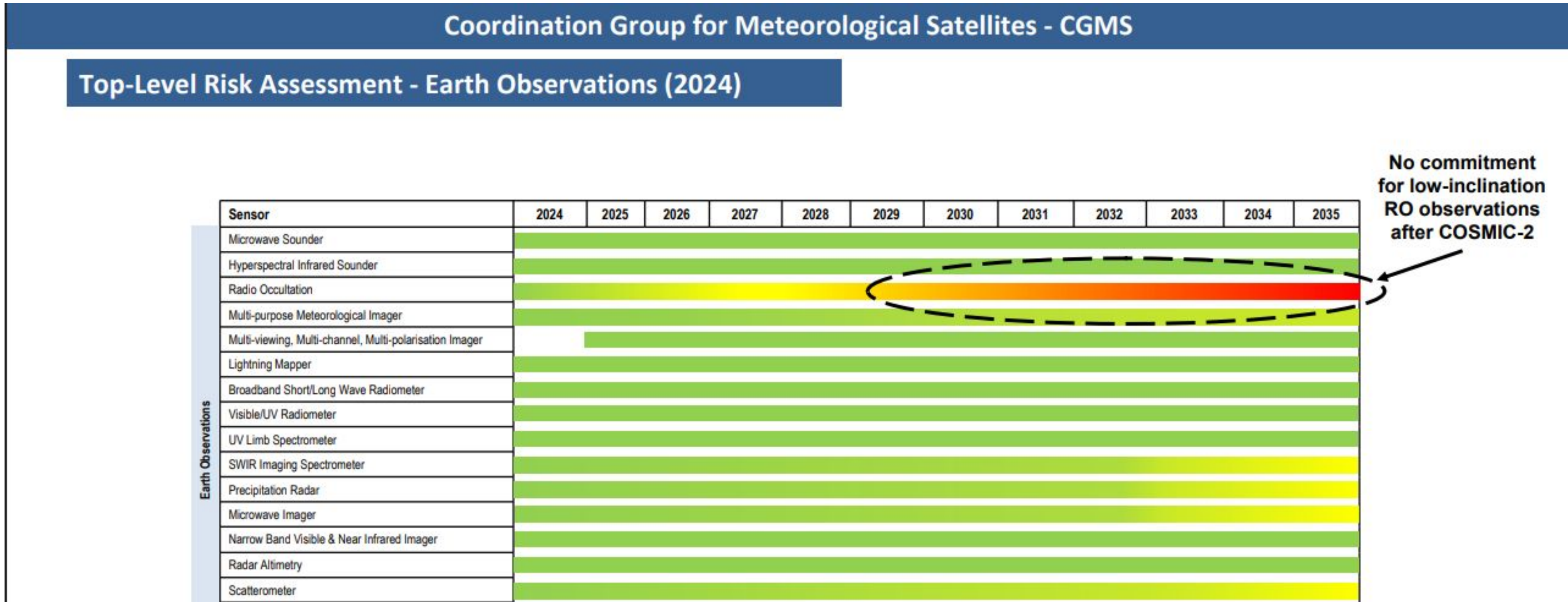


Microwave Sounding

NOAA plans to procure commercial data to supplement the government backbone, particularly beyond QuickSounder in the 17:30 orbit.

Equatorial RO

As announced in last year, NOAA plans to procure commercial data to maintain critical continuity as the COSMIC-2 satellites approach end of operational lifetimes.



Source: CGMS WGIII 6th risk assessment, 22 April 2024


THANK YOU!

NOAA
4/09/2026

NOAA Acquisition & Grants Office (AGO)

Professional, Scientific, and Technical Services (ProTech)
Commercial Space Based Environmental Monitoring (SBEM) Domain

Michael Conroy, Director, Corporate Services Acquisition Division

ProTech Overview

- ProTech is a strategically sourced portfolio of mission aligned domains contracts to provide scientific and other technical capabilities to all NOAA Line and Staff Offices
- Current domains include
  - Satellite
  - Weather
  - Oceans
  - Fisheries
- The domain contracts are 10 year Multiple Award Indefinite Delivery/Indefinite Quantity contracts solicited as small business set asides
- The base contracts are administered through the ProTech Program Office
- Individual orders are placed on a decentralized basis through the four operational contracting offices within NOAA
- The shared program ceiling is $8B

ProTech Program Strategy

- Each domain consists of a contracting officer and an account manager who also serves as the contracting officers representative for the base contracts

- Alignment of acquisition programs with programmatic areas

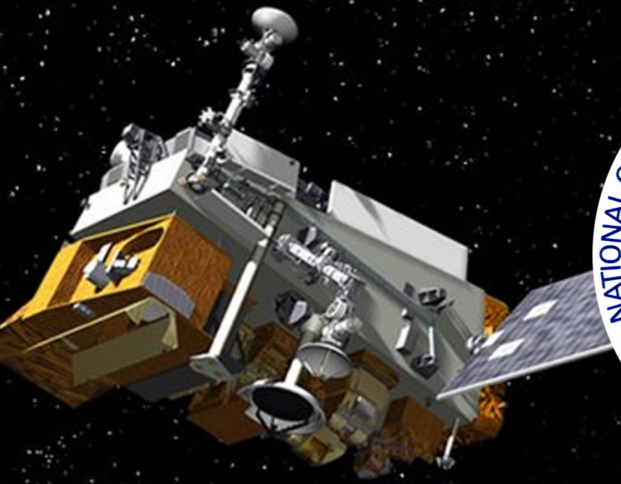
- Build strong partnership with industry
  - Continual industry engagement
  - Acquisition forecasting and planning
  - Annual technical capability updates
  - Onboard of newly capable vendors

Space Based Environmental Monitoring Domain
(General Acquisition Strategy)

- Multiple Award Indefinite Delivery/Indefinite Quantity Contract solicited on a full and open basis with a reserve for small business
- 10-yr period of performance with a five year base and 5 year option
- Continuous onboarding for newly capable vendors
- Estimated value (with options) of $2.3B of the $8B shared ceiling
- Primarily for NESDIS but available to other NOAA offices if within the contract scope
- Until all approvals are received and the final solicitation issued all acquisition strategy is subject to change

Summary

- Thank You for your interest in supporting NOAAs Mission!
- Points of Contact:
  - Michael Conroy: Michael.conroy@noaa.gov
  - ProTech Branch Chief: David Marks David.marks@noaa.gov
  - Contracting Officer: Paul Batrony Paul.Batrony@noaa.gov
  - Contract Specialist: Daniel Ok Daniel.ok@noaa.gov



National Environmental Satellite, Data, and Information Service

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NOAA NESDIS Commercial Data Program Industry Day

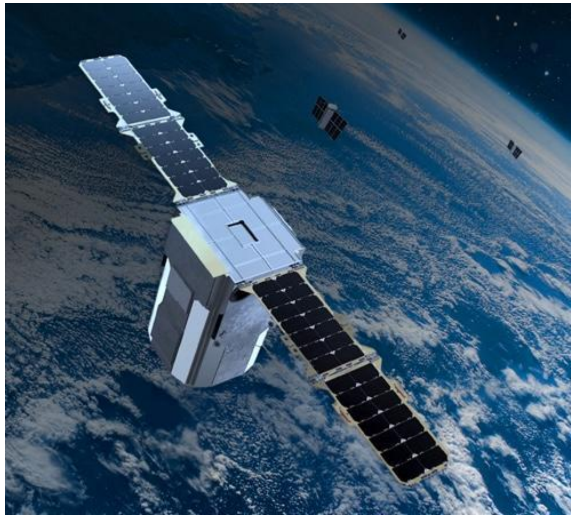
Natalie Laudier, Commercial Data Program Manager

NOAA NESDIS Systems Architecture and Engineering

NESDIS Commercial Data Program Supports NOAA's Strategy

NOAA pursues a hybrid architecture including procurement of space-based commercial data:

- *Government owned systems*
- *Commercial data*
- *International partnerships*



Artist representation of a Spire SmallSat constellation. Credit: Spire

"Buy and Partner where we can, Build what we must."

Commercial weather data improves coverage and enables more ***accurate weather forecasts***.

NESDIS Commercial Data Program (CDP) Mission

PURPOSE : Acquire commercial space-based environmental observation data to diversify, streamline, and improve NOAA's observations and to meet NOAA’s mission objectives.

TWO PILLARS :

1. **Commercial Data Pilots:**
   Demonstrates the quality and impact of commercial data on NOAA applications

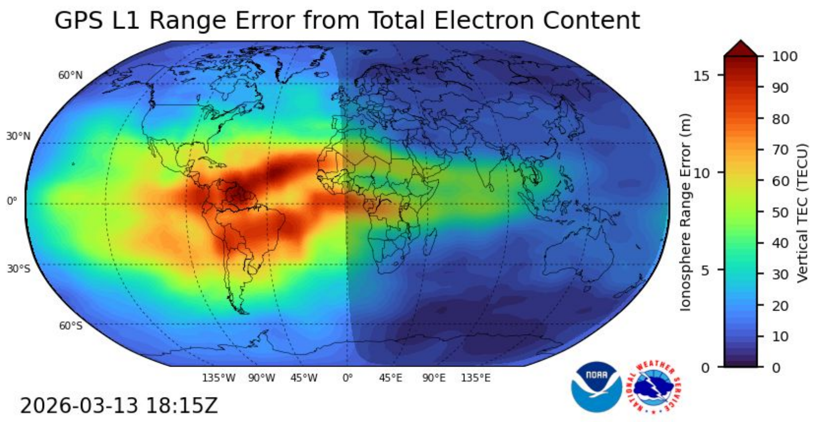
2. **Commercial Data Purchases:**
   Procures data-as-a-service from commercial vendors for operational weather forecasting and space environment applications

IMPACT: Improves the accuracy and timeliness of NOAA’s weather forecasts, space weather and atmospheric monitoring, ocean observations, and other critical environmental applications.

Commercial Data Types

Successful Transition to Operations (T2O):

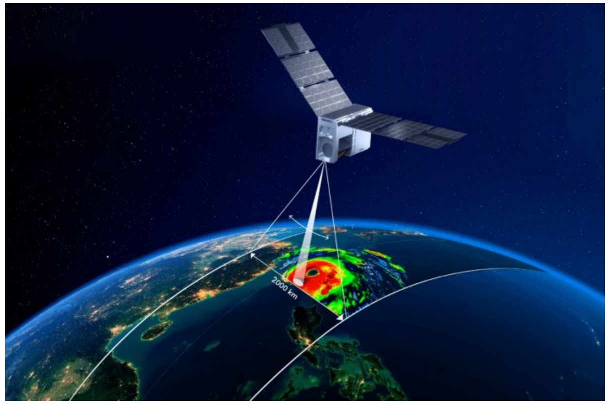
- Radio Occultation (RO)
- Total Electron Content



GloTEC Total Electron Content (TEC) (ground + COSMIC-2 + PlanetIQ)

Current/FY26 Piloting:

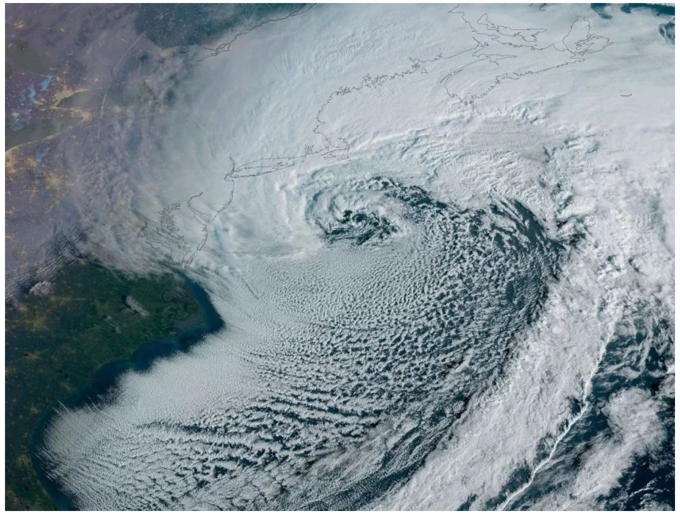
- GNSS-Reflectometry
- Microwave Sounder- including Hyperspectral
- IR Imagery (for Wildfires)



Tomorrow.io Microwave Sounder rendering

Future Considerations (not limited to):

- Thermospheric Neutral Density
- Scintillation (RO-derived)
- Scatterometry
- Multispectral Imagery
- IR Sounders



2026 Northeast Blizzard GOES Visible Image

Process Strategy for Transition to Ops (T2O)

1. Requirements & Investment Drivers
2. Pilot Data Evaluation, Operational Value & User Feedback
3. Utility Determination
4. Data Integration into NOAA Operational Systems
5. Operational Readiness and Sustainment

NESDIS Data Centers

Cooperative Institutes

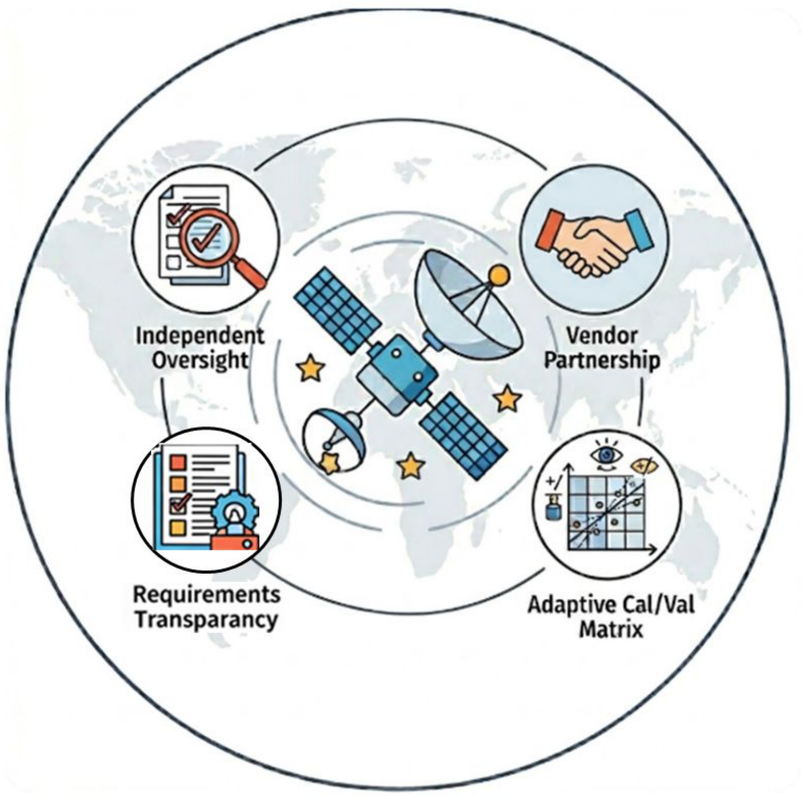
NESDIS Flight Programs

NESDIS CDP ensures full operational integration of commercial data throughout the transition-to-operations process.

Commercial Satellite Data Calibration/Validation Strategy

Cal/Val is the foundation for reliable, decision-ready commercial satellite data. NESDIS CDP is establishing an agile, standardized, enterprise-level process to facilitate transitioning commercial data to operational assets.

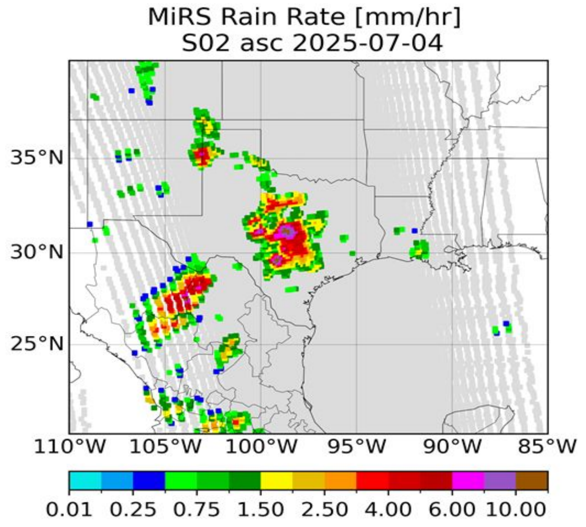
- **Establish Adaptive Cal/Val Matrix:** Define a framework that maps purchased commercial data quality to maturity level.
- **Increase Requirements Transparency:** Set clear specifications to guide vendors and incentivize prioritization of data quality.
- **Guarantee Independent Oversight:** Maintain objective QC, Cal/Val, and performance assessments through dedicated government teams.
- **Enhance Vendor Partnership:** Foster collaboration by providing commercial partners with clear standards and performance metrics.

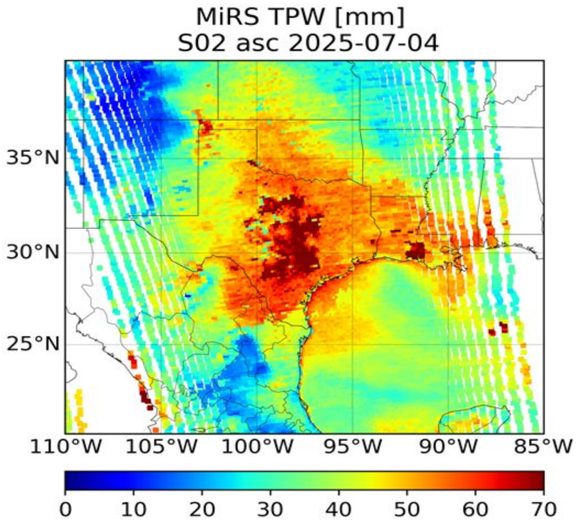


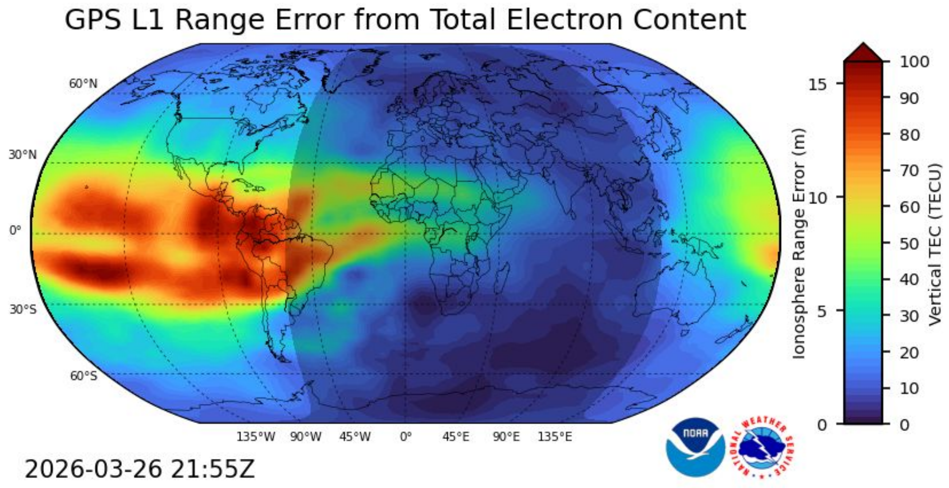
Credit: Gemini AI

2025 NESDIS CDP Accomplishments

- **Successfully transitioned Total Electron Content (TEC) data to Operations in Delivery Order 5**
  - Awarded first oper. purchase of Space Wx; SWPC validating data in Global TEC model via the SWPC Testbed
- **Completed GNSS-RO Analysis of Alternatives (AoA)**
  - Concluded study to optimize RO architectural solutions for the best value to the Government
- **Completed GNSS-R Pilot for Ocean Surface Winds and additional GNSS-R products**
  - Initiated follow-on Pilot to facilitate transition to operational purchasing
- **Executed 12-month data delivery for the Microwave Sounder Pilot**







Credits: NOAA

NESDIS CDP Operational Radio Occultation Data Buy (RODB)

NESDIS CDP successfully purchases and integrates commercial GNSS-RO data for operations.

Delivery Order 5 (Sep 2025 – Sep 2026)

- 10,000 RO profiles/day (split between vendors)
- 2,500 Total Electron Content (TEC) tracks/day
- Total Cost: $35.5M

PlanetiQ

6,500 RO profiles/day

500 High SNR RO profiles/day

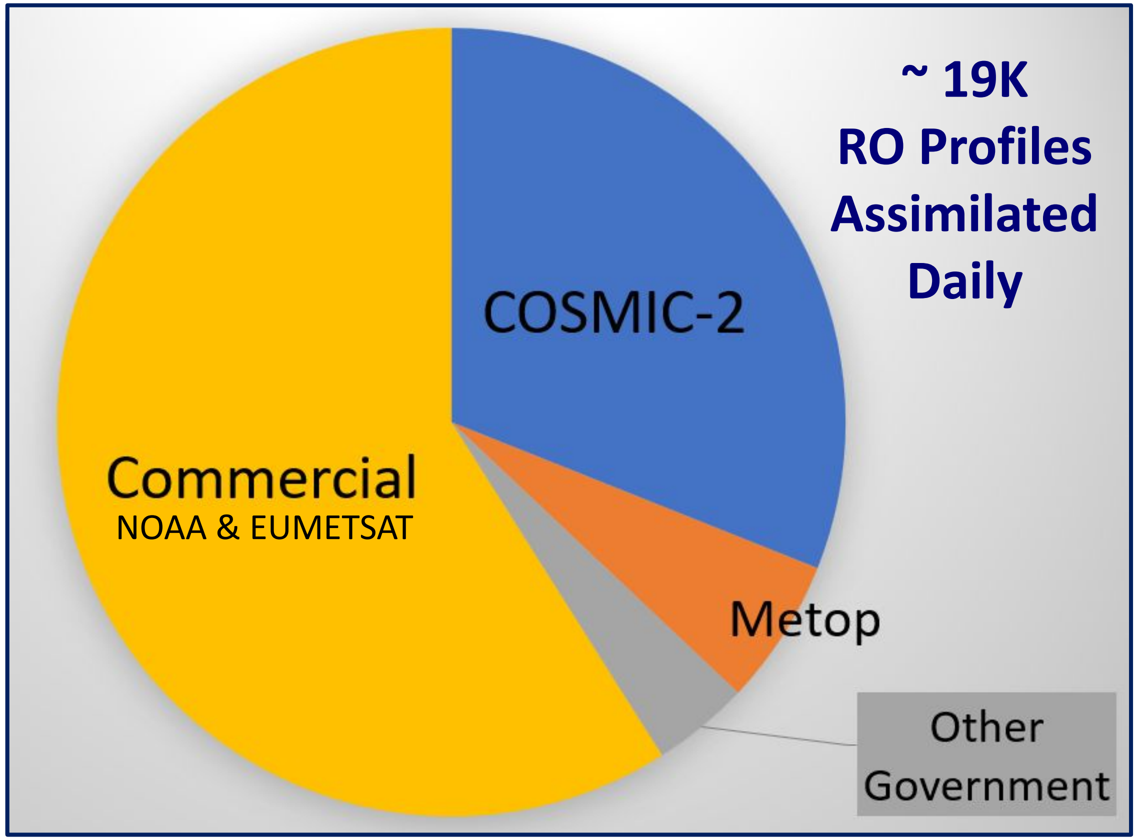
2,500 TEC tracks/day

Spire Global

3,000 RO profiles/day

DO-6 (Sep 18 to Dec 1, 2026)

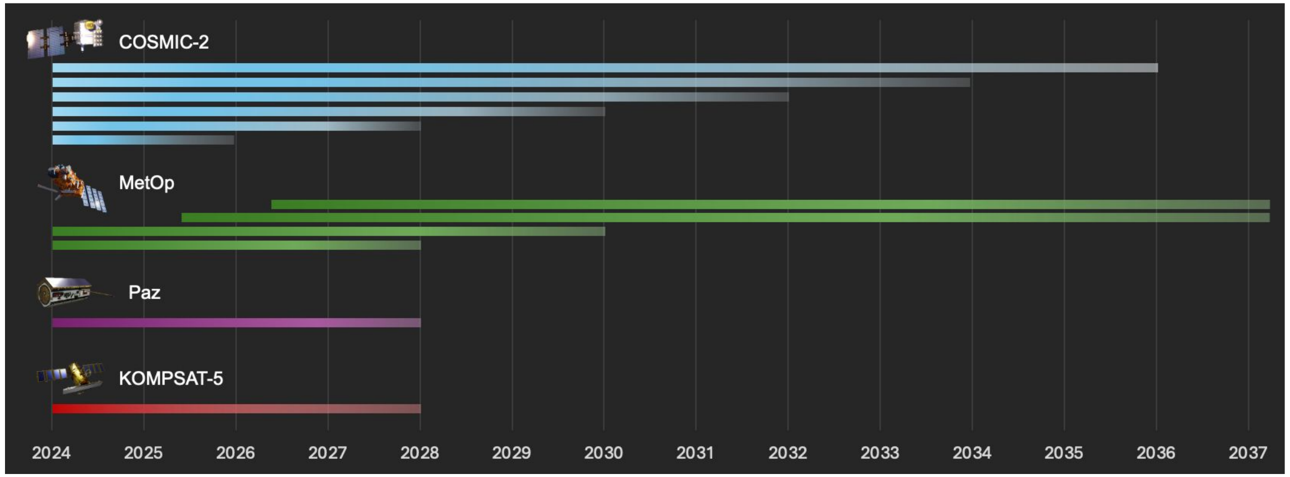
Bridge to SBEM TO-1 (begins Dec 1)

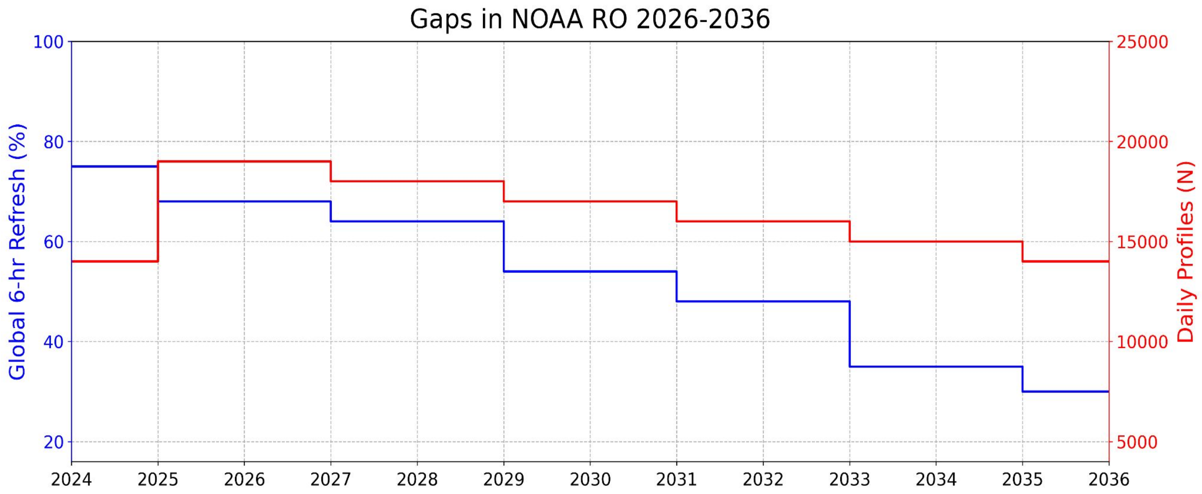


RO Data assimilated into NWS NWP models
(as of Dec 2025)

NOAA/NESDIS Approach to Radio Occultation (RO)

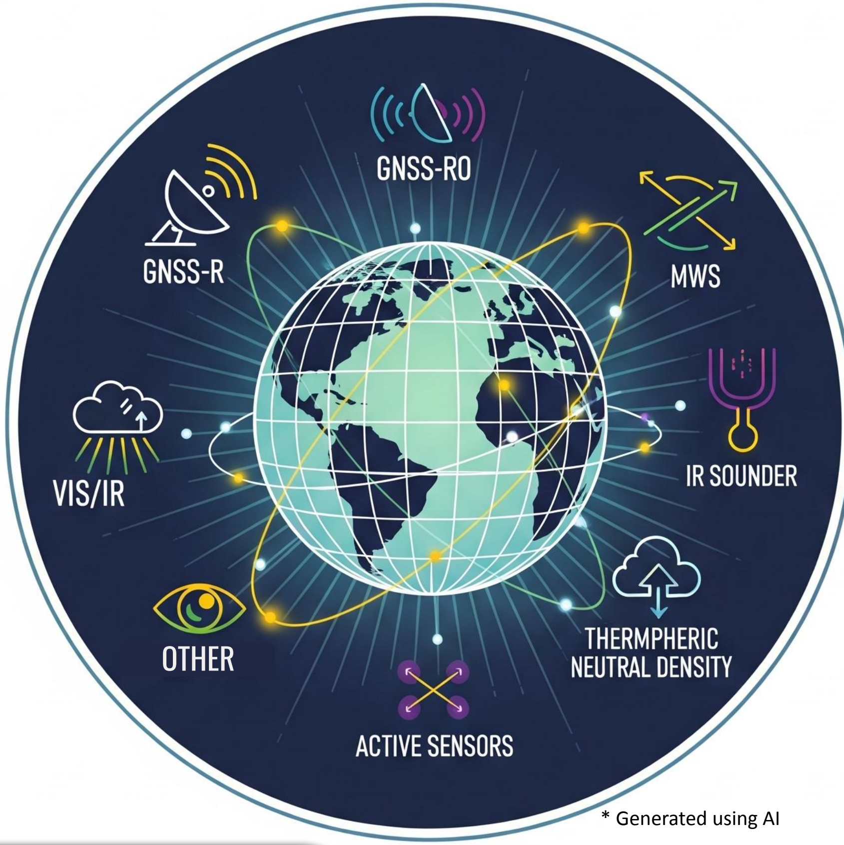
- NOAA/NESDIS goal for Radio Occultation is a Hybrid Architecture (Current and Future)
  - Commercial Data
  - International Partnership
- NOAA is committed to increase RO commercial data purchases globally - for equatorial, polar, and mid-latitude regions.
- NOAA conducted an **RO Analysis of Alternatives (AoA)** Study on potential solutions to fill gaps as **COSMIC-2 degrades** and proposed solutions to maintain capability to meet NOAA's RO requirements through 2036.





The Space-Based Environmental Monitoring (SBEM) Indefinite Delivery Indefinite Quantity (IDIQ)

- New Domain under ProTech IDIQ called SBEM for the purchase of commercial environmental satellite data
- Will include various data types from space-based instruments sensing environmental parameters (Functional Categories)
- Allows for both Piloting and Operational Purchases
- DaaS model: Only award for data delivered
- Continuous onboarding for new vendors and vendors with new capabilities
- On-Orbit requirements expanded
    - ✓ Current
    - ✓ Planned (NEW)

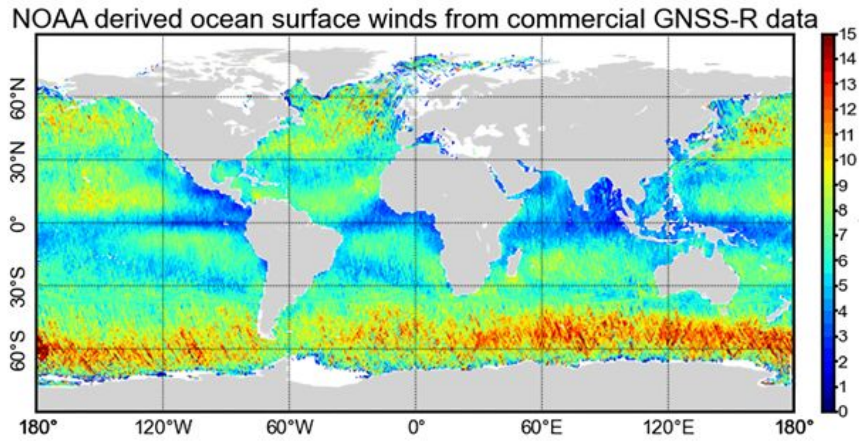


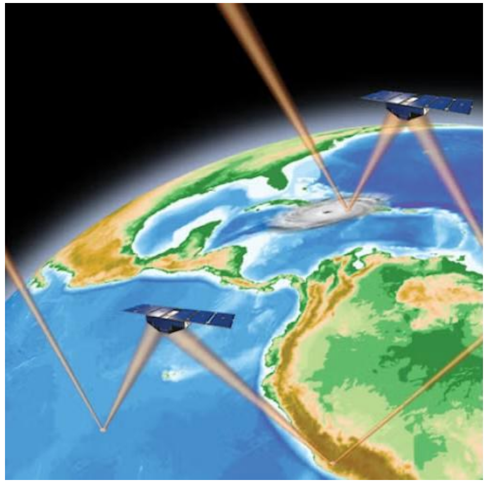
* Generated using AI

Target Award	Release Request for Proposals	PoP
End of FY 2026	April 2026	10 Years

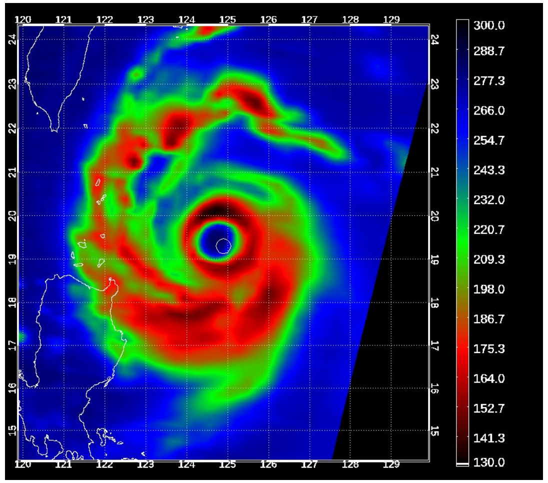
NESDIS Commercial Data Pilots

- **GNSS-Reflectometry Ocean Surface Winds (OSW) Follow On Pilot** – *Ongoing*
- **Microwave Sounder Pilot** – *Ongoing*
- **Microwave Sounder Data Buy** – *2026*
- **Wildfire Imagery Pilot** – *2026*

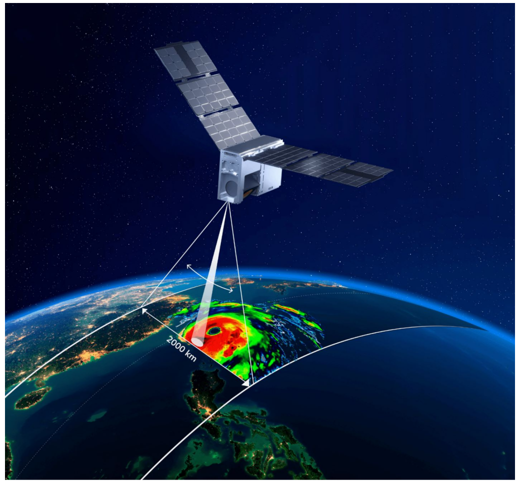




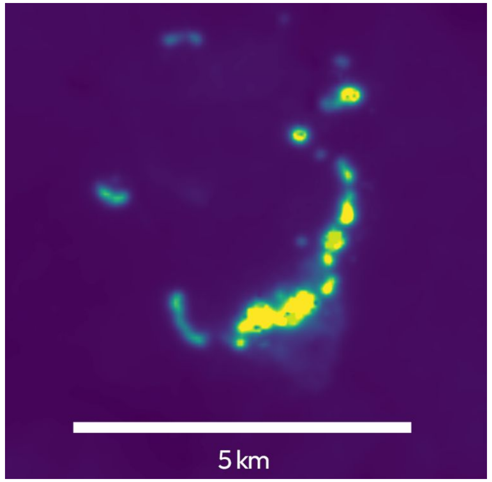
Credit: UCAR



Credit: CIMMS/NOAA



Credit: Tomorrow.io



Credit: Muon Space

The NESDIS CDP Pilot Process

MARKET RESEARCH

PILOT PROPOSAL SOLICITATION

PROPOSAL EVALUATION & PILOT AWARD

PILOT EXECUTION

TRANSITION TO OPERATIONS

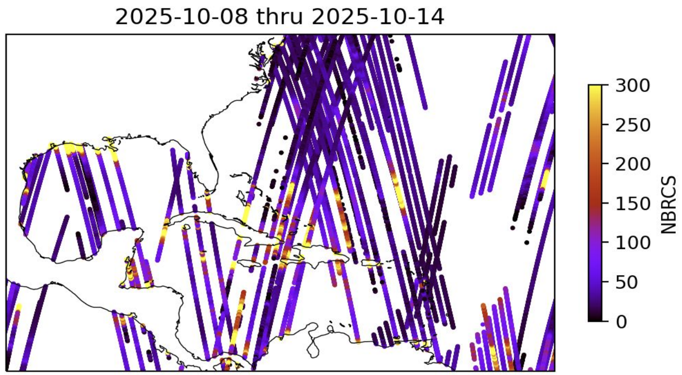
Are we ready for a Pilot?

Are we ready for Operational Use?

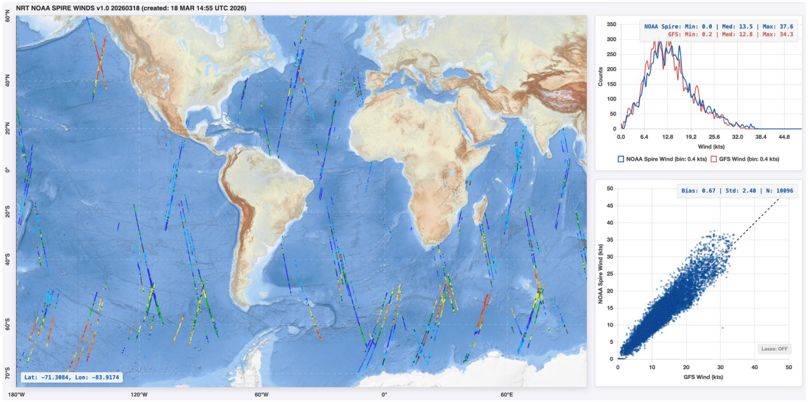
GNSS-Reflectometry (GNSS-R) Ocean Surface Winds II Pilot

Follow-on to GNSS-R Ocean Surface Winds (OSW) Pilot

- **Purpose**: Evaluate commercial GNSS-R data to derive ocean surface wind speeds for use in Numerical Weather Prediction (NWP) and Tropical Cyclone (TC) prediction and monitoring
- **Pilot Goals**:
  - Evaluate potential of new high Signal to Noise Ratio (SNR) sensors for better OSW characterization and hurricane prediction
  - Further assess impact of GNSS-R OSW data on NWP
  - Assess new data for use in storm and TC prediction and monitoring
- **Status:**
  - Awarded to Muon Space and Spire Global
  - Development of NOAA Commercial GNSS-R Ocean Surface Winds Product
  - 6-month data delivery period ended in March '26
  - In Evaluation Phase with vendor support, NWS EMC and JCSDA evaluation of OSW product



7-day aggregate of specular reflections from CDP Pilot Muon GNSS-R data. The Normalized Bistatic Radar Cross-Section represents the scattering coefficient of the surface: high-magnitude reflections over the ocean indicate changes in sea surface roughness driven by local wind speeds. Credit: Spire Global



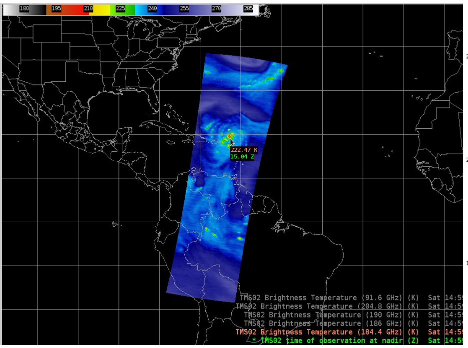
Single-day aggregate of derived Ocean Surface Wind Speeds from CDP Pilot Spire GNSS-R data and correlation to GFS model wind speeds. Credit: NOAA STAR

Microwave Sounder (MWS) Pilot

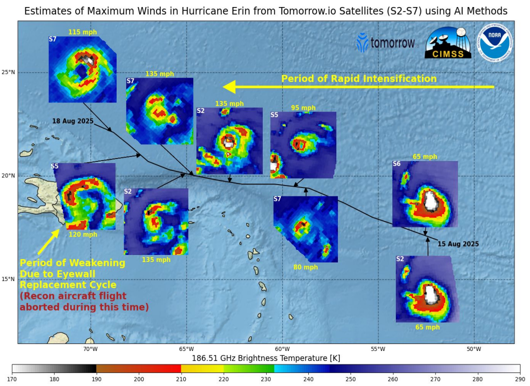
NOAA successfully awarded a Microwave Sounder Pilot to Tomorrow.io

- **Purpose**: Evaluate commercial MWS data to enhance NWP and improve forecaster situational awareness
- **Status**:
  - Initial demonstrations of Tropical Cyclone (TC) imagery shows good capability - positive feedback from NHC.
  - Identified vendor constellation performance limitations.
  - Ongoing data assimilation for model forecast impacts.
  - 3-month data extension (through May 2026) fully supports L2 product evaluation and additional TC imagery validation.
  - Transition to Ops development is underway. CDP working with stakeholders to standardize cal/val practices.
- **Milestones**:
  - June 2026: Final pilot results complete (notional)
  - ***June 2026: Next MWS commercial data buy begins. RFP was released on April 2, 2026, proposals due April 22.***

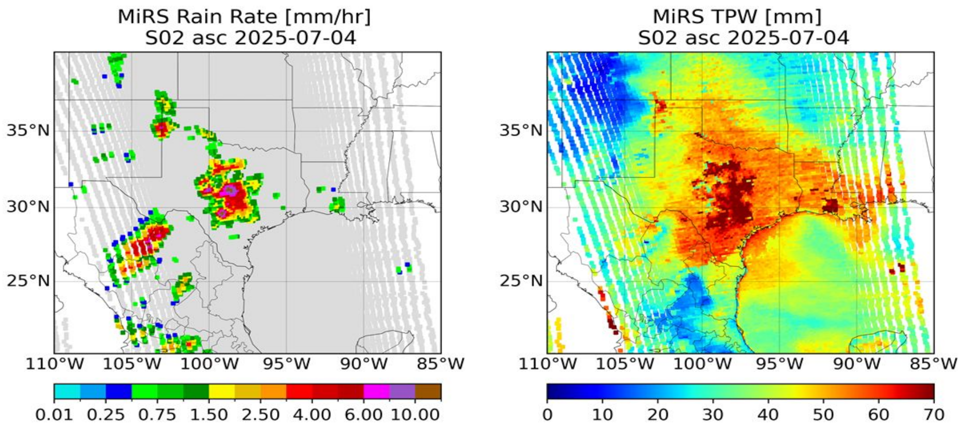
Tropic Cyclone imagery in AWIPS2



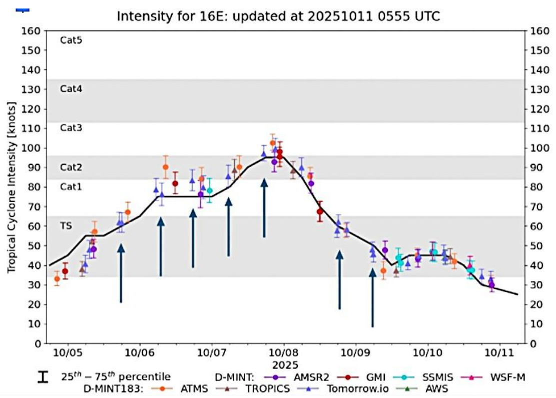
Tropical Intensity AI Demonstration



Rain Rate and TPW retrieved from Tomorrow.io TMS-02



Hurricane Intensity Estimates



2026 Wildfire Imagery Pilot

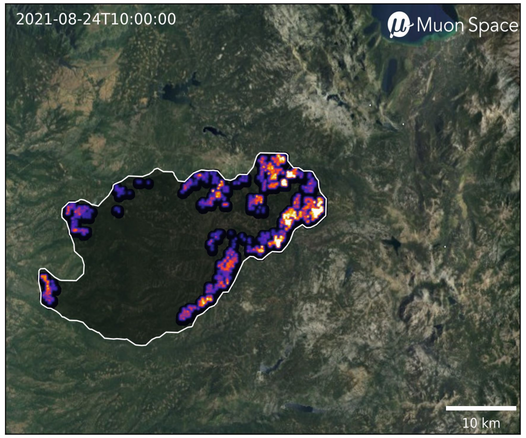
Purpose: Assess high-resolution multispectral IR imagery for wildfire decision support

Pilot Goals:

- Advance product development in 3 main areas:
  - Early detection: Identify wildfires when they are small and more manageable
  - Fire intensity monitoring: Monitor radiative power of fires and track smoke production
  - Fire mapping: Precisely geolocate active fire regions to include fire intensity mapping
- Assess quality, value, utility, and tech readiness relative to critical fire weather mission needs
- Assess the viability of assimilating commercially-provided satellite data and products and evaluate their value within an integrated data and application framework
- Leverage NWS Forecast Offices, on-site incident meteorologists and test bed to assess vendor data

Status:

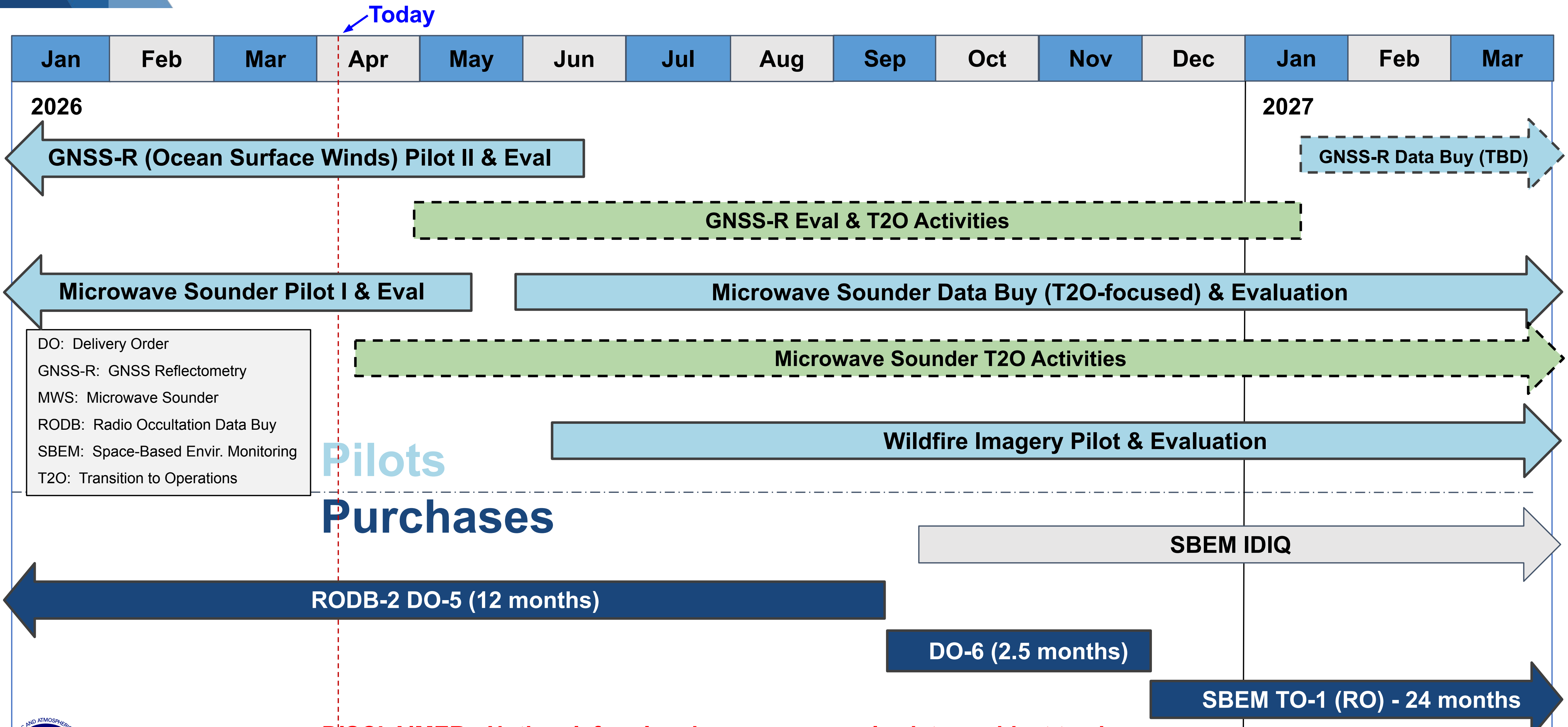
- CDP released a draft SOW for public review in February detailing data requirements; vendor community feedback was received and used to inform changes to the SOW.
- **The RFP was released on Sam.gov on April 7th; proposals are due by April 30, 2026.**



Fire radiative power (simulated).
Credit: Muon Space

Angeles National Forest firefighters at the Eaton Fire, Jan. 21, 2025.
Credit: USDA Forest Service

2026 - 2027 NESDIS CDP Planning - Pilots & Purchases



DISCLAIMER: Notional, for planning purposes only, dates subject to change

Questions?

NESDIS Commercial Data Program Information:

https://www.nesdis.noaa.gov/commercial-data-program

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